

ECONOMETRICS I

Changes and Errata

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PROBLEM SETS

Problem Set 2

- (3a) Slight changes in the labels of the graph.
- (3b) **Important change.** In an early version, the condition $\mathbb{E}[\text{cov}(x, u)] = 0$ was erroneously stated without the expectation operator $\mathbb{E}()$.
- (4a) By the end of the answer, I expanded for clarity:

$$\begin{aligned}\text{var}(\hat{\beta}_2|x) &= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \mathbb{E} \left[\left(\sum_{i=1}^n (x_i - \bar{x}) u_i \right)^2 \middle| x \right] \\ &= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\end{aligned}$$

- (4c) **Important change.** Previously I wrote this nonsense (perhaps copy-pasting from somewhere): Recall that under $\mathbb{E}(X'u) = 0$ it holds that

$$\mathbb{E}(\hat{\beta}) = \beta + (X'X)^{-1}X'u \quad (3)$$

The corrected sentence now reads: Recall that, assuming $y = X\beta + u$, we get

$$\hat{\beta} = (X'X)^{-1}X'y = \beta + (X'X)^{-1}X'u, \quad (3)$$

and that, assuming $\mathbb{E}(X'u) = 0$, we get

$$\mathbb{E}(\hat{\beta}|X) = \beta. \quad (4)$$

Problem set 6

- **Important.** Question 2 was changed entirely. It is now auto-contained and offers a good exercise to understand when IV estimates work and when they don't.
- Some redundant questions were erased or slightly changed.

LECTURES

Lecture 3

- Change made a month ago: New description of R^2

Lecture 10

- **Important.** Slide 25, point 2 said “uncorrelated with u” but should be “uncorrelated with $\mathbf{u}$ ”. I also changed “cov” to better fit the matrix notation: $E(\mathbf{X}'\mathbf{u})$ instead of $\text{cov}(\mathbf{X}, \mathbf{u})$.